# 16. Singular matrices

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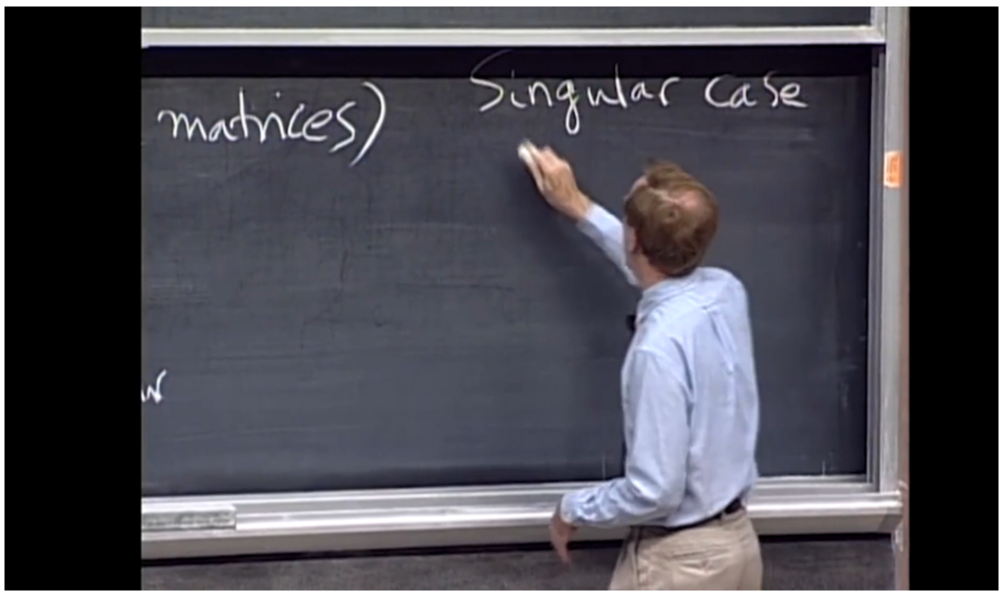
Lecture due Feb 14, 2024 20:30 IST Completed

# Singular matrices

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**Can I talk about the singular case, no inverse?**

All right.

Best to start with an example.

Tell me an example, let's get an example up here.

Let's make it two by two matrix that has not got an inverse.

And let's see why.

Let me write one up.

No inverse

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Matrices that have no inverse are called **singular.** The theorem for invertible matrices can be stated in terms of the complementary conditions for singular matrices. (It's essentially the same theorem, so this is really just a review.)

**Theorem 16.1** For a square matrix $\mathbf{A}$, the following are equivalent:

1. $\det \mathbf{A} = 0$
2. $\mathrm{NS}\left(\mathbf{A}\right)$ is larger than $\{\mathbf{0}\}$ (i.e., $\mathbf{Ax} = \mathbf{0}$ has a nonzero solution)
3. $\mathbf{rank}\left(\mathbf{A}\right) < n$ (image has dimension less than $n$)
4. $\mathrm{CS}\left(\mathbf{A}\right)$ is smaller than $\mathbb{R}^n$ (image is not the whole space $\mathbb{R}^n$)
5. The system $\mathbf{Ax} = \mathbf{b}$ has no solutions for some vectors $\mathbf{b}$, and infinitely many solutions for other vectors $\mathbf{b}$.
6. $\mathbf{A}^{-1}$ does not exist.
7. $\mathtt{rref}\left(\mathbf{A}\right) \neq \mathbf{I}$

## Singular matrices concept check

1/1 point (graded)

Suppose $\mathbf{A}$ is any matrix, and $\mathbf{A}$, $\mathbf{B}$, and $\mathbf{C}$ are three $n \times n$ matrices. **True or False?** If $\mathbf{AB} = \mathbf{AC}$, then $\mathbf{B} = \mathbf{C}$.

- [ ] True
- [x] False

**Solution:**

False. Suppose that $\mathbf{A}$ is the zero matrix, then $\mathbf{AB} = \mathbf{AC}$ for all $n \times n$ matrices $\mathbf{B}$ and $\mathbf{C}$. (Moreover, if $\det \mathbf{A} = 0$ we can always find distinct $\mathbf{B}$ and $\mathbf{C}$ such that $\mathbf{AB} = \mathbf{AC}$. As a challenge problem, try to construct such matrices $\mathbf{B}$ and $\mathbf{C}$ given a singular matrix $\mathbf{A}$.) If $\mathbf{A}$ is invertible, however; then multiplying by $\mathbf{A}^{-1}$ we see that $\mathbf{B} = \mathbf{C}$.

This problem demonstrates one of the many properties true of invertible matrices, but not of matrices in general.

Submit You have used 1 of 1 attempt

Answers are displayed within the problem

Calculator

Now let's explain the consequences of $\det \mathbf{A} = 0$.

1. The input space is flattened by $\mathbf{A}$. This means that some input dimensions are getting crushed, so $\mathrm{NS}\left(\mathbf{A}\right)$ is larger than $\{\mathbf{0}\}$ (at least $1$-dimensional), and the image is smaller than the $n$-dimensional input space: $\mathbf{rank}\left(\mathbf{A}\right) < n$.

2. In particular, the image $\mathrm{CS}\left(\mathbf{A}\right)$ is not all of $\mathbb{R}^n$.

3. If $\mathbf{b}$ is not in $\mathrm{CS}\left(\mathbf{A}\right)$, then $\mathbf{Ax} = \mathbf{b}$ has no solution.

4. If $\mathbf{b}$ is in $\mathrm{CS}\left(\mathbf{A}\right)$, then $\mathbf{Ax} = \mathbf{b}$ has the same number of solutions as $\mathbf{Ax} = \mathbf{0}$, i.e., infinitely many since $\dim \mathrm{NS}\left(\mathbf{A}\right) \geq 1$.

5. The associated linear transformation $\mathbf{f}$ is not a $1$-to-$1$ correspondence (it maps many vectors to $\mathbf{0}$); thus $\mathbf{f}^{-1}$ does not exist, so $\mathbf{A}^{-1}$ does not exist. Row operations do not change the condition $\det \mathbf{A} = 0$, so $\det\left(\mathtt{rref}\left(\mathbf{A}\right)\right) = 0$, so definitely $\mathtt{rref}\left(\mathbf{A}\right) \neq \mathbf{I}$. (In fact, $\mathtt{rref}\left(A\right)$ must have at least one $0$ along the diagonal.)

**Problem 16.2** Devise a test for deciding whether a homogeneous square system $\mathbf{Ax} = \mathbf{0}$ has a nonzero solution.

Solution

Compute $\det \mathbf{A}$. If $\det \mathbf{A} = 0$, there exists a nonzero solution. If $\det \mathbf{A} \neq 0$, then $\mathbf{Ax} = \mathbf{0}$ has only the zero solution.

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## 16. Singular matrices

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